Changkai Chen

Astronomy Ph.D. student, Beijing Normal University
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Education

Beijing Normal University
Ph.D. in Astronomy · 2025–present
Supervisor: Prof. Zhoujian Cao. Expected June 2027.
Radiation fluxes for gravitational self-force modelling and black hole quasinormal mode spectra [1,2,7,8]. Current work: preparing manuscripts on Reissner–Nordström and Kerr–Newman spectra; developing efficient time domain codes for extreme mass ratio inspirals.

Research Assistant, Hunan Normal University
2024–2025.

Hunan Normal University
M.Sc. in Astronomy · 2021–2024
Supervisor: Prof. Jiliang Jing.
Derived connection formulas for confluent Heun functions to determine scattering amplitudes, extending radiation calculations from circular Schwarzschild orbits to generic Kerr motion [3,4].

China Three Gorges University
B.Sc. in Physics · 2016–2020
Developed compact finite differences and precise time integration for fluid dynamics and parabolic equations [5,6].

Research Interests

Black hole perturbation theory; gravitational wave astronomy; gravitational self-force and inspiral modelling.

Publications

Six first-author peer-reviewed articles.

[1] Changkai Chen, Zhoujian Cao, and Jiliang Jing. Efficient and stable computation of gravitational-wave fluxes from generic Kerr orbits via a unified Heun-function framework. Phys. Rev. D 114, 063002 (2026).

[2] Changkai Chen, Jiliang Jing, Zhoujian Cao, and Mengjie Wang. Complete quasinormal modes of type-D black holes. Phys. Rev. D 112, 103036 (2025).

[3] Changkai Chen and Jiliang Jing. Gravitational wave fluxes on generic orbits in near-extreme Kerr spacetime: Higher spin and large eccentricity. Sci. China Phys. Mech. Astron. 67, 110411 (2024).

[4] Changkai Chen and Jiliang Jing. Radiation fluxes of gravitational, electromagnetic, and scalar perturbations in type-D black holes: an exact approach. JCAP 11 (2023) 070.

[5] Changkai Chen, Xiaohua Zhang, and Zhang Liu. A high-order compact finite difference scheme and precise integration method based on modified Hopf-Cole transformation for numerical simulation of n-dimensional Burgers' system. Appl. Math. Comput. 372, 125009 (2020).

[6] Changkai Chen, Xiaohua Zhang, Zhang Liu, and Yage Zhang. A new high-order compact finite difference scheme based on precise integration method for the numerical simulation of parabolic equations. Adv. Differ. Equ. 2020, 15 (2020).

Preprints

Two first-author preprints.

[7] Changkai Chen, Xiaohua Zhang, Zhoujian Cao, Jiliang Jing, and Sheng Long. Complete total-transmission modes of Kerr black holes. arXiv:2609.39609 [gr-qc] (2026), preprint.

[8] Changkai Chen, Xiaohua Zhang, Zhoujian Cao, and Jiliang Jing. Complete quasinormal modes of Type-D black holes. II. Extremal Kerr black holes. arXiv:2610.39095 [gr-qc] (2026), preprint.

Research Software and Data

Complete-QNMs. Schwarzschild and Kerr quasinormal mode (QNM) spectra: numerical data and visualization code [2].

Complete-TTMs. Kerr total-transmission modes (TTMs): spectral datasets, root-finding examples and visualization code [7].

GWFluxHeunC. Gravitational wave energy fluxes from generic Kerr orbits, providing energy-balance data for orbit-averaged dissipative gravitational self-force (GSF) calculations [1].

CFD-PIM. Undergraduate computational fluid dynamics: compact finite differences and precise time integration for one-, two- and three-dimensional Burgers systems [5].

Selected Talks and Posters

Complete Spectroscopy of Kerr Black Holes

1 Nov 2026. Hangzhou Workshop on Gravitation and Cosmology, Hangzhou.

Complete Quasinormal-Mode Spectra of Black Holes

1 Jun 2026. Invited seminar, Peking University. Host: Prof. Xian Chen.

Black Hole Modes and Gravitational-Wave Flux in Loop Quantum Gravity

26 May 2026. Loops'26 International Conference on Quantum Gravity, Hangzhou.

Complete Spectroscopy of Black Holes

19 Apr 2026. Annual Meeting on Gravitation and Relativistic Astrophysics, Chinese Physical Society, Haikou. Also presented the poster *Solving Black Hole "Ringing" Mysteries*.

Exact Wave-Optical Imaging and Complete Spectroscopy of Black Holes

22 Dec 2025. Black Hole Imaging Workshop, Yangzhou University.

Conferences and Training

6th Symposium of the BRICS Association on Gravity, Astrophysics, and Cosmology

6–10 Nov 2026. Participant, Hangzhou.

Adaptive Finite-Element Algorithms and AFEPack Applications

20 Jul–4 Aug 2026. Training course, Yunnan Normal University, Kunming.

Honours

CAST Young Science and Technology Talent Development Program

2026. Selected for the Doctoral Student Special Program, China Association for Science and Technology.

Outstanding Presentation Award

2026. Annual Meeting on Gravitation and Relativistic Astrophysics, Chinese Physical Society.